The American Journal of Clinical Nutrition 124 (2026) 101346

American Society for Nutrition
Excellence in Nutrition Research and Practice

The American Journal of
CLINICAL NUTRITION

journal homepage: https://ajcn.nutrition.org/

Original Research Article

Krill oil increases plasma omega-3 fatty acids more than fish oil in healthy adults: a double-blind randomized controlled trial

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A B S T R A C T

Background: Omega-3 fatty acids (ω-3 FAs), particularly EPA and DHA, are recognized for their health benefits. However, their circulating levels after supplementation may be modulated by several factors, including sex, carriage of the apolipoprotein E ε4 (APOE4) allele, and the chemical form of the supplement. Krill oil delivers ω-3 FAs primarily as phospholipids, whereas fish oil provides them as triglycerides.

Objectives: This study aimed to compare EPA and DHA concentrations after supplementation with krill oil and fish oil and assess whether sex and APOE4 genotype modify responses to supplementation.

Methods: This double-blind, randomized clinical trial included 72 healthy adults (53 females, 19 males) matched for age and BMI. Participants received 1.1 g/d ω-3 FAs through either krill oil ($n = 36$) or fish oil ($n = 36$) for 12 wk. Plasma FAs were measured at baseline and at weeks 1, 2, 4, and 12 by GC–flame ionization detection. Differences in plasma ω-3 FAs concentrations by treatment, sex, and APOE4 status were analyzed.

Results: Time-by-treatment interactions were significant for plasma delta over baseline concentrations of EPA ($P = 0.0001$) and DHA ($P = 0.005$), with krill oil resulting in ~1.5-fold higher Δ EPA and Δ DHA compared with fish oil. The time-by-sex interaction was significant only for EPA ($P = 0.026$), with females having a 1.5-fold greater increase than males at 12 wk. After supplementation with either krill oil or fish oil, APOE4 carriers had 3-fold and 1.6-fold higher EPA and DHA, respectively, compared with baseline; however, these increases were not significantly different from those found in noncarriers.

Conclusions: Krill oil increased plasma ω-3 FAs more than fish oil, regardless of APOE4 genotype. Individuals with higher ω-3 FA requirements may achieve adequate enrichment with lower doses of krill oil compared with fish oil supplementation.

This trial was registered at clinicaltrials.gov as NCT04279743.

Keywords: EPA, DHA, supplementation, Krill oil, phospholipids, RCT, Alzheimer's, APOE4

Introduction

Omega-3 fatty acids (ω-3 FAs) are widely recognized for their health benefits, particularly for cardiovascular [1,2], metabolic [3,4], brain structure [5], and function [6–8]. Among them, EPA and DHA have been associated with a reduced risk of cognitive decline and late-onset Alzheimer's disease (LOAD), defined by clinical onset after 65 y of age [9–11]. However, a recent review by our group found that only 44% of randomized controlled trials (RCTs) reported cognitive benefits after ω-3 FAs supplementation [12]. These inconsistencies may partly be explained by the presence of the apolipoprotein E ε4 (APOE4) allele, the main genetic risk factor for LOAD. Our team previously reported a gene-by-diet interaction, where APOE4 carriers had a lower increase in ω-3 FAs compared with noncarriers after supplementation [13]. Moreover, sex might modulate the response to ω-3 FAs supplementation, with females having higher increases in EPA and DHA than males after supplementation [14].

The chemical form of ω-3 FAs can also modulate the response to supplementation in plasma [15–18] and tissues [19,20]. A recent review emphasized that different sources and esterification forms of EPA

Abbreviations: ANOVA, analysis of variance; APOE4, apolipoprotein E ε4; DPA, docosapentaenoic acid; FFA, free fatty acid; FFQ, Food Frequency Questionnaire; FID, flame ionization detector; LOAD, late-onset Alzheimer's disease; LPC, lysophosphatidylcholine; ω-3 FA, omega-3 fatty acid; PC, phosphatidylcholine; PL, phospholipids; RCT, randomized controlled trial; TG, triglycerides.

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https://doi.org/10.1016/j.ajcnut.2026.101346
Received 31 October 2025; Received in revised form 4 May 2026; Accepted 8 May 2026; Available online 16 May 2026

and DHA can lead to distinct biomarker responses, including total plasma lipids and the ω-3 index [21]. Consistently, a study comparing fish-oil–derived ω-3 FAs provided as triglycerides (TG) and ethyl esters, with krill oil in which EPA and DHA are primarily esterified in phospholipids (PL), reported the highest incorporation of ω-3 FAs into plasma PL with krill oil [18]. Although krill oil is often associated with higher EPA and DHA concentrations, findings from human trials remain inconsistent. Most studies were limited by short supplementation periods [18,22,23], small sample sizes with sex bias [18,24,25], unequal capsule numbers [26,27], or unmatched EPA and DHA doses [25,28]. Only 1 double-blind RCT using matched EPA and DHA doses reported higher plasma ω-3 FAs with krill oil. However, the fish-oil supplement contained higher ω-6 FAs levels, potentially biasing the comparison between treatments [17].

The provision of PL-ω-3 FAs is of particular interest due to its potential to enhance transport of DHA to the brain by facilitating transport across the blood-brain barrier [29]. Therefore, we sought to compare similar doses of EPA and DHA differing only by their esterification form: TG (fish oil) compared with PL (krill oil) while accounting for sex and APOE4 genotype. We hypothesized that krill oil increases plasma EPA and DHA more than fish oil and that plasma ω-3 FAs concentrations are modulated by APOE4 genotype and sex. The primary outcome was the change from baseline in EPA and DHA concentrations after 12 wk of supplementation with krill oil and fish oil. The secondary outcomes were plasma EPA and DHA in APOE4 carriers compared with non-carriers and in males compared with females with fish and krill supplements.

Methods

Participants and study design

This study was a double-blind, RCT registered on clinicaltrials.gov (NCT04279743) on 21 February, 2020. All participants provided written informed consent, and the study was approved by the Human Ethics Research Committee of the Health and Social Sciences Center—Sherbrooke University Geriatrics Institute (IRSC2020-3570). Healthy males and females aged 30 to 65 y were recruited from the general population and were matched for age and BMI. This age range was selected because the study was designed as a metabolic investigation aimed at informing preventive strategies for LOAD. We, therefore, focused on healthy adults, before the emergence of age-related neurodegenerative or metabolic changes, to characterize ω-3 fatty acid metabolism under physiologically stable conditions. Exclusion criteria included current smoking, consumption of oily fish more than twice per week, use of ω-3 FAs supplements, or lipid-lowering medications. Individuals were also excluded if they presented with malnutrition (assessed via serum albumin, BMI <18.5 kg/m^2, etc.), diabetes mellitus, or uncontrolled thyroid, renal, hepatic, or endocrine disorders. Additional exclusions applied to participants with chronic inflammatory or immune conditions (defined as C-reactive protein >10 mg/L or abnormal white blood cell counts), a history of cancer, recent major surgery or cardiac events, or those who were pregnant, lactating, or perimenopausal. Participants were also excluded if they had a diagnosis of dementia or self-reported any neurodegenerative disorder. No formal cognitive assessment was conducted as part of the eligibility criteria. Participants were also excluded if they had a history of severe substance abuse, major psychiatric disorders (e.g., major depression), or if they were currently or previously engaged in intensive physical training.

Participant recruitment and follow-up visits were conducted between April 2022 and November 2024. Among the 305 individuals screened for eligibility, 205 were excluded based on exclusion criteria (n = 127), genotyping (n = 52), or blood test results (n = 26). Of the 100 randomly assigned participants, 24 participants declined before starting the supplementation, leaving 38 randomly assigned participants in each supplement group. There were 2 dropouts from each supplement group. Reasons for dropout included difficulty swallowing capsules (n = 2), digestive discomfort (n = 1), and inability to attend study visits (n = 1).

The supplements used in this study were obtained from 2 different production batches for both krill oil and fish oil. Detailed information regarding manufacturing lots, reception dates, declared shelf life, and EPA and DHA concentrations is provided in Supplementary Table 1. Fish oil capsules were obtained from Neptune Wellness Solutions, and krill oil capsules (SuperbaBoost) were provided by Aker BioMarine Antarctic AS. Participants consumed 4 capsules of 1000 mg daily for 12 wk, 2 with breakfast and 2 with dinner. The 4 capsules provided a total of ~1100 mg of ω-3 FAs daily. Specifically, the krill oil group received 706 mg EPA and 393 mg DHA, whereas the fish oil group received 738 mg EPA and 462 mg DHA per day. The detailed fatty acid composition of each supplement, as analyzed by our laboratory, is provided in Table 1. Between-lot variability was calculated as the absolute percentage difference relative to the mean concentration of the fatty acid in the 2 production batches:

% variability = |Concentration in lot1 − Concentration in lot2 | / [(Concentration in lot1 + Concentration in lot2)/2] × 100. The percentage of variability between the 2 lots is provided in Table 1. EPA and DHA concentrations showed <15% variability between production batches for both supplements.

Krill oil provided ω-3 FAs predominantly esterified in PL, with PL accounting for 57% of total lipids, of which 90% were in the form of

TABLE 1
Fatty acid composition in milligrams per gram of capsule

Fatty acid	Krill capsules	Fish capsules	% Variability	
			Krill	Fish
C12:0	1.16 ± 0.01	1.00 ± 0.03	4.40	35.53
C13:0	0.38 ± 0.59	ND	51.37	ND
C14:0	60.70 ± 0.36	75.24 ± 0.57	13.99	15.98
C14:1	1.03 ± 0.02	0.29 ± 0.41	7.66	200
C15:0	2.49 ± 0.03	5.15 ± 0.03	8.78	10.34
C15:1	ND	0.49 ± 0.30	ND	50.93
C16:0	173.35 ± 0.97	172.96 ± 2.47	2.69	10.75
C16:1 t	1.08 ± 0.07	3.77 ± 0.09	1.46	9.01
C16:1	27.70 ± 0.17	68.12 ± 35.15	1.32	50.47
C17:0	ND	5.59 ± 0.18	ND	2.97
C17:1	1.02 ± 0.01	1.44 ± 0.08	4.60	27.63
C18:0	4.46 ± 0.04	19.20 ± 0.24	148.05	186.22
C18:1 ω-9	70.73 ± 0.35	115.13 ± 1.59	0.42	6.92
C18:1 ω-7	47.70 ± 0.36	31.79 ± 0.46	7.66	3.33
C18:2 t	1.77 ± 0.14	0.43 ± 0.00	18.99	200
C18:2 ω-6	12.03 ± 0.04	10.87 ± 0.08	41.90	30.90
C18:3 ω-6	0.95 ± 0.52	2.58 ± 0.04	50.88	1.02
C18:3 ω-3	24.27 ± 0.09	7.93 ± 0.06	22.63	4.32
C20:0	ND	2.98 ± 0.06	ND	77.07
C20:1	3.85 ± 0.03	12.92 ± 0.19	10.61	35.94
C20:4 ω-6	1.50 ± 0.06	9.50 ± 0.12	200	31.05
C20:5 ω-3	176.57 ± 1.14	184.56 ± 2.22	3.15	**4.74**
C22:5 ω-6	ND	1.71 ± 0.02	ND	200
C22:5 ω-3	4.01 ± 0.15	19.78 ± 0.24	19.41	5.83
C22:6 ω-3	98.20 ± 0.48	115.45 ± 1.16	**15.68**	**0.79**

Abbreviation: ND, not detected; ω, omega.

phosphatidylcholine (PC) (Supplementary Table 1). The remaining lipid fraction may be in other esterified forms, such as TG or free fatty acids (FFAs), which were not further characterized by manufacturers. Each krill oil capsule delivered 70 mg choline, corresponding to a daily intake of 280 mg choline. In contrast, ω-3 FAs from fish oil were primarily esterified in TG.

At baseline visit, body height and weight were measured at the metabolic unit of the research center on aging in Sherbrooke, using a calibrated scale, and BMI was calculated as weight (kg)/height (m^2). Blood samples were collected in EDTA-coated tubes at baseline and then at 1, 2, 4, and 12 wk of supplementation (Supplementary Figure 1). Blood was then centrifuged at 2300 × g for 15 min at 4°C and afterward, plasma was aliquoted and stored at −80°C 7]. Plasma samples from each participant's visits were then allocated a random number to maintain the blinding from the treatment and time point during the analysis. Blood biochemistry was performed at baseline and included measurements of fasting glucose, total cholesterol, TG, HDL cholesterol, and LDL cholesterol.

APOE genotyping

At the screening visit, 5 mL of venous blood was collected from each participant into EDTA-coated tubes and stored at −80°C (Supplementary Figure 1). Genotyping was conducted at the RNomics Platform of the University of Sherbrooke. Genomic DNA was extracted from thawed whole blood using the QIAamp DNA Extraction Kit (Qiagen), following the manufacturer's instructions, as described by Maltais et al. [30]. Participants were classified into 2 genetic groups based on the presence of the APOE4 allele. Only participants with the APOE3/APOE3 genotype (APOE4 noncarriers) and APOE3/APOE4 genotype (APOE4 carriers) were included in the study. The research team remained blind to participants' APOE genotypes throughout the recruitment and sample processing phases.

Randomization and blinding

All research staff were blinded to both treatment allocations until the end of study recruitment. The file linking participant names and personal information to their randomization numbers was deleted once recruitment was completed.

The random allocation sequence (krill oil and fish oil) was generated using a computer-based random number generator by the principal investigator, who was not involved in participant enrolment. Participant recruitment and enrolment were conducted by a research nurse and a PhD student who were blinded to both the allocation sequence and genotypes. Randomization was stratified by sex and APOE genotype, and participants were matched for age and BMI to ensure balanced distribution between treatment groups.

The principal investigator had access to the randomization codes and genotype information, but did not have access to any file containing participants' personal identifiers. Supplement capsules were provided in identical, coded containers labeled as "treatment 1" and "treatment 2," and participants remained blinded to their treatment assignment throughout the study.

Plasma lipid extraction

42 μg of TG heptadecanoic acid (C17:0) was added to 100 μL plasma samples as an internal standard. Total lipids were extracted using a chloroform: methanol mixture (2:1, v/v) as previously described [31]. The lipid extracts were saponified by incubation with 3 mL of potassium hydroxide in methanol Potassium hydroxyde in methanol (KOH-MeOH) (56 g/L) at 90°C for 60 min. The saponification mixture was then acidified using 0.3 mL of 12 M hydrochloric acid. FFAs were methylated with 3 mL of 12% boron trifluoride in methanol Boron trifuoride in methanol (BF3-MeOH) (Thermo Scientific Chemicals) at 90°C for 30 min. The resulting methyl esters were reconstituted in 500 μL of hexane and transferred into GC vials.

Fatty acids methyl esters were analyzed using GC (model 6890; Agilent Technologies) equipped with a BPX-70 fused silica capillary column (50 m; SGE). Detection was carried out with a flame ionization detector (FID), following the parameters described in Chevalier et al. [16]. Plasma fatty acids methyl esters were analyzed as single determinations. Prior validation of the method in our laboratory has shown reproducibility with an SD <6% for plasma fatty acids. The chromatogram analysis was performed using the OpenLab CDS ChemStation software. Quantification was based on the internal standard and calibration curves constructed using relative response factors derived from a certified fatty acid methyl esters standard mixture (Nu Chek Prep GLC-569B). The GC-FID used in this study had a limit of detection of 0.21 to 0.54 μg/mL and corresponding limits of quantification of ~0.63 to 1.63 μg/mL. Fatty acids with concentrations below the limit of detection (LOD) were considered not detected and are reported as "ND."

Food frequency questionnaire

Dietary intake of ω-3 FAs was assessed at the baseline visit using a validated Canadian food frequency questionnaire (FFQ) specifically designed to capture foods naturally rich in or enriched with ω-3 FAs [32]. The questionnaire was adapted to include locally available food brands and names in Québec, Canada. The FFQ was translated into French by 2 certified translators. To ensure the accuracy and consistency of the translated content, a back-translation into English was subsequently performed by a third certified translator. This process verified the fidelity of the French version to the original English FFQ (the 2 FFQs in French and English can be found in the Supplementary File). Dietary intake of EPA and DHA was then determined from FFQs with the use of The Food Processor SQL Edition dietary analysis software (version 11.15.1, 2025, ESHA Research).

Participants were instructed to maintain their usual daily routines, including eating habits and physical activity levels, throughout the study period. They were also asked to limit their intake of fatty fish to no >2 servings per week and to moderate alcohol consumption. Each participant was provided with a logbook to record their consumption of marine-based foods during the intervention.

Physical activity assessment

Physical activity was assessed using ActiGraph GT3X accelerometers. At baseline, participants were instructed to wear the device over their dominant hip (i.e., right hip for right-handed individuals and left hip for left-handed individuals) during waking hours for 7 consecutive days, excluding water-based activities. The accelerometers captured habitual physical activity during the first week of supplementation. Data were recorded in 10-s epochs and analyzed using ActiLife software (ActiGraph LLC), applying the Freedson Adult (1998) count cut-points for activity classification as follows: sedentary (0–99 counts/min), light (100–1951 counts/min), moderate (1952–5724 counts/min), vigorous (5725–9498 counts/min), and very vigorous (≥9499 counts/min). A valid day was defined as ≤10 h of wear time, and participants were included in the analysis if they recorded a minimum of 4 valid days [33]. Data from 66 participants (33 per group of supplementations) met these inclusion criteria, whereas 7 participants were excluded due to noncompliance with wear instructions or failure to achieve the minimum wear-time requirements.

Statistical analysis

The sample size was calculated using the change in ω-3 FAs after 4 wk supplementation with krill oil as compared with fish oil from the study by Ramprasath et al. [17]. The calculated effect size (Cohen's $d = 0.686$) was used to determine the sample size required to achieve 80% statistical power and a significance level (α) of 0.05. A minimum of 33 participants per group was required to meet these statistical criteria. This sample size estimation was based on the primary comparison between supplements.

Anthropometric characteristics were assessed using unpaired *t*-tests for normally distributed data or Mann–Whitney *U* tests for nonnormally distributed data. On the basis of pill count, adherence to the supplementation was calculated for each participant as a percentage. Adherence was compared between groups using an unpaired *t*-test. For the primary outcome, a 2-way analysis of variance (ANOVA) was performed to evaluate the main effects of treatment (krill oil as compared with fish oil) and time (baseline, 1, 2, 4, and 12 wk of supplementation), as well as their interaction on the change from baseline (Δ) in EPA and DHA. For secondary outcomes, a 3-way ANOVA was conducted to evaluate the effects of treatment, time, and additional factors (sex and APOE4 status), including their interactions, on plasma ΔEPA and ΔDHA concentrations. EPA and DHA Δ concentrations were used to calculate the AUC to assess total plasma exposure to the supplements.

Missing data were not imputed, and analyses were conducted on available data only. Statistical significance was set at $P < 0.05$. All statistical analyses were performed using GraphPad Prism version 9.3.1 for Windows (GraphPad Software). We used the CONSORT reporting guideline [34] to draft this manuscript, and the CONSORT reporting checklist [34] when editing, included in supplementary data.

Results

Study population

A total of 72 participants completed the trial, with 36 individuals in each intervention group (fish oil and krill oil) (Figure 1). Each group included 10 APOE4 carriers and 26 noncarriers, with a comparable sex distribution (26 females and 10 males in the fish oil group; 27 females and 9 males in the krill oil group) (Figure 1). The overall study population consisted of 53 females and 19 males, with a median age of 59 y (41, 63) and a median BMI of 25.72 kg/m^2 (22.79, 28.59). Baseline characteristics of participants are presented in Table 2. All blood biochemistry data, including glucose and lipid parameters, were within the clinical normal range. Participants had an average of 31 min of moderate-to-vigorous physical activity per day. Possible side effects were assessed at each study visit by asking participants about any adverse symptoms or discomfort related to supplement use. No side effects were reported by participants in either supplementation group. At baseline, participants reported a median dietary intake of total EPA+DHA of 210 mg/d (102, 332.5), with no significant differences in EPA, DHA, or total EPA+DHA intake between the 2 supplementation groups (Table 2).

Compliance with ω-3 FAs supplementation

On the basis of the capsule count, median (Q1, Q3) compliance to fish oil was 96% (90%, 99%) and 97% (91%, 99%) for those receiving the krill oil. There was no significant difference in compliance between the 2 groups ($P = 0.314$). Minimum compliance was 62 % ($n = 1$) in the fish oil group and 79% ($n = 1$) in the krill oil group. Plasma ω-3 FAs concentrations after 12 wk of supplementation were ≥1.2-fold higher than baseline, such that no participant was excluded from the dataset.

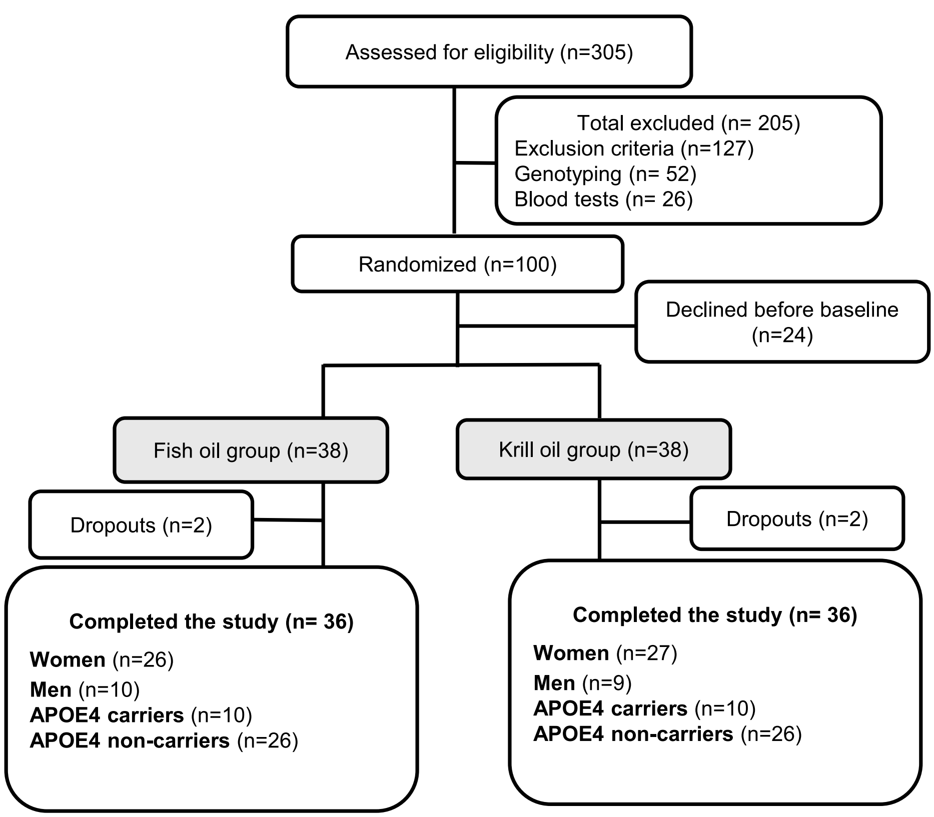

FIGURE 1. Participants recruitment flowchart. APOE4, apolipoprotein E ε4.

TABLE 2
Clinical characteristics of participants at baseline

	Total	Fish group (n = 36)	Krill group (n = 36)
Age (y)	59 (41, 63)	58.5 (41.5, 63)	59 (41, 63)
Females, *n* (%)	53 (73.6)	26 (72.2)	27 (75)
Ethnicity, *n* (%)			
Others	3 (4.2)	2 (5.6)	1 (2.8)
APOE4 carriers, n (%)	20 (27.7)	10 (27.7)	10 (27.7)
BMI (kg/m^2)	25.72 (22.79, 28.59)	26.21 (22.75, 28.8)	25.3 (22.77, 28.1)
Glucose (mmol/L)	5.38 ± 0.56	5.39 ± 0.54	5.38 ± 0.58
Triglycerides (mmol/L)	0.92 (0.65, 1.26)	0.93 (0.64, 1.26)	0.81 (0.69, 1.31)
Total cholesterol (mmol/L)	5.14 ± 1.08	5.07 ± 1.14	5.21 ± 1.03
HDL cholesterol (mmol/L)	1.64 (1.25, 1.87)	1.5 (1.23, 1.75)	1.73 (1.37, 1.91)
LDL cholesterol (mmol/L)	2.95 (2.51, 3.6)	2.97 (2.51, 3.6)	2.95 (2.48, 3.61)
MVPA /d (min)	30.98 ± 20.55	29.48 ± 21.63	32.48 ± 19.97
Dietary ω-3 FAs			
EPA (mg/d)	70 (40, 120)	60 (30, 120)	80 (50, 100)
DHA (mg/d)	140 (60, 220)	125 (60, 235)	150 (70,190)
EPA + DHA (mg/d)	210 (102, 332.5)	190 (85, 357)	230 (120, 290)

Results are means ± SD when data were normally distributed and median (Q1, Q3) when not normally distributed. Means between groups were compared using an unpaired *t*-test when normally distributed, and medians were compared using a Mann–Whitney nonparametric test when data were not normally distributed.
Abbreviations: APOE4, apolipoprotein E ε4; FA, fatty acid; HDL-C, MVPA, moderate-to-vigorous physical activity; ω-3 FA, omega-3 fatty acid.

Plasma concentrations of EPA and DHA in plasma

To evaluate the increase in EPA and DHA levels during supplementation, baseline plasma concentrations were subtracted from those measured at subsequent visits. The resulting values are referred to as ΔEPA and ΔDHA, representing the net change in plasma levels of these fatty acids over time. There was a significant time × supplement interaction for both plasma ΔEPA (Figure 2A) and ΔDHA (Figure 2B) ($P < 0.05$), with higher plasma concentrations in participants supplemented with krill oil compared with fish oil. The ΔEPA and ΔDHA followed different patterns, with plasma ΔEPA increasing sharply within the first week of supplementation, after rising more gradually between weeks 1 and 12. At week 12, the ΔEPA was significantly higher in the krill oil group (ΔEPA = 6.88 ± 0.39 mg/dL) compared with the fish oil group (ΔEPA = 4.90 ± 0.40 mg/dL). For DHA, Δconcentrations increased between baseline and 4 wk and then reached a plateau between 4 and 12 wk (ΔDHA = 4.22 ± 0.31 mg/dL) for krill oil, whereas for fish oil, plasma concentrations increased between baseline and 2 wk and then reached a plateau between 2 and 12 wk of supplementation (ΔDHA = 2.84 ± 0.34mg/dL). Overall, supplementation with krill oil resulted in a 1.4-fold higher ΔEPA and a 1.5-fold higher ΔDHA compared with fish oil supplementation at 12 wk. The total plasma fatty acids profile at baseline and 12 wk by supplement is reported in Supplementary Table 2.

Plasma concentrations of EPA and DHA by sex

There was a significant time × sex interaction for plasma ΔEPA (Figure 3A, $P = 0.026$), suggesting sex differences in the increases of ΔEPA concentrations over time. Females had significantly higher increases in plasma ΔEPA concentrations than males, regardless of supplement type (krill, fish) (Figure 3A, P sex < 0.001). In contrast, no significant time × sex interaction was found for plasma ΔDHA concentrations (Figure 3B), suggesting similar DHA increases over time in both sexes. Krill oil–supplemented females had 1.42- and 1.62-fold higher plasma ΔEPA than females in fish oil and men in krill oil, respectively, at 12 wk of supplementation. However, females in the Krill-oil group had 1.53- and 1.37-fold higher ΔDHA than females in the fish oil group and males in the krill oil group, respectively, at 12 wk (Figure 3B). Absolute concentrations of EPA and DHA at baseline and at 12 wk of supplementation in men and females are reported in Supplementary Figure 2A, B. Total plasma fatty acids profile at baseline and 12 wk by sex are reported in Supplementary Table 3.

Plasma concentrations of EPA and DHA by APOE genotype

There were no significant genotype × time or genotype × supplement interactions for plasma ΔEPA and ΔDHA (Figure 4A, B), indicating that the effect of supplementation over time was similar

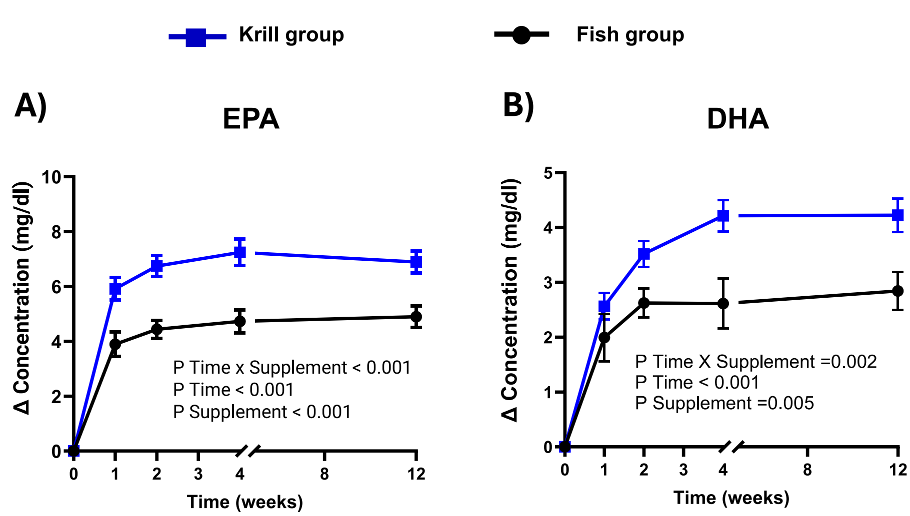

FIGURE 2. Plasma delta concentrations from baseline of (A) EPA and (B) DHA with Krill-oil supplement (blue) and Fish-oil supplement (black). Values are means ± SEM. Missing data were as follows: in week 4 (fish oil, n = 2); week 12 (fish oil, n = 1; krill oil, n = 1).

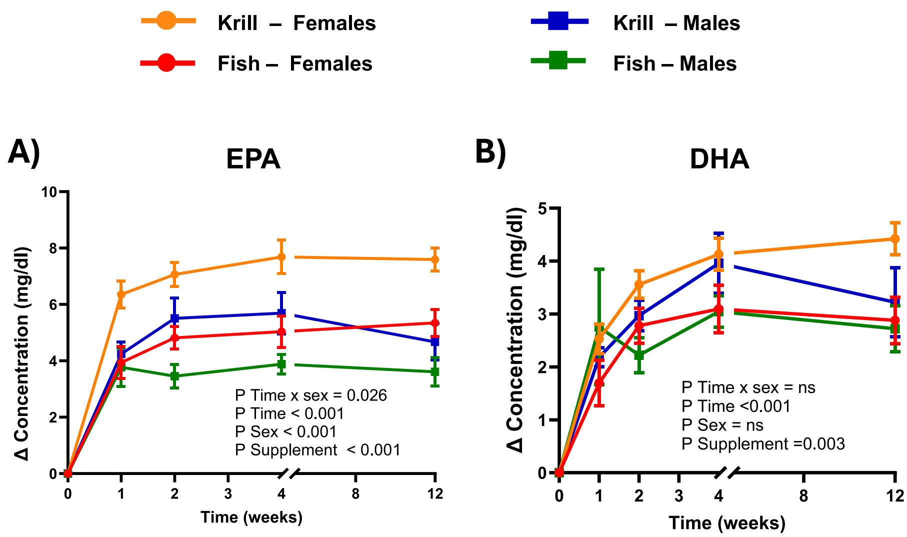

FIGURE 3. Plasma delta concentrations from baseline of (A) EPA and (B) DHA with Krill-oil supplement (orange; females, blue; males) and Fish-oil supplement (red; females, green; males). Missing data were as follows: at week 4 (females, fish oil, $n = 1$; males, fish oil, $n = 1$) and at week 12 (females, krill oil, $n = 1$; males, fish oil, $n = 1$).

regardless of APOE genotype. However, plasma ΔEPA and ΔDHA concentrations increased significantly over time ($P < 0.001$), with krill oil leading to greater increases than fish oil ($P < 0.001$). Therefore, both APOE4 carriers and noncarriers (APOE3) had similar increases in ΔEPA and ΔDHA over time, with higher increases in ΔEPA and ΔDHA with krill oil as compared with APOE4 carriers and noncarriers supplemented with fish oil. Absolute plasma concentrations at baseline and at 12 wk by APOE genotype are reported in Supplementary Figure 2C, D, and total plasma fatty acids profile at baseline and 12 wk by APOE4 carriers and noncarriers are reported in Supplementary Table 4. APOE4 carriers had 3-fold higher EPA and 1.60-fold higher DHA at 12 wk as compared with baseline ($P < 0.001$); however, this was not statistically different from APOE4 noncarriers (Supplementary Figure 2C, D).

Discussion

The present study tested the hypothesis that krill oil (PL ω-3 FAs) increases plasma EPA and DHA more than fish oil (TG ω-3 FAs). To address this, we evaluated whether a 12-wk supplementation with a matched dose of ω-3 FAs from krill oil would lead to higher increases in plasma EPA and DHA compared with fish oil in healthy males and females. We found that krill oil supplementation led to ~1.5-fold higher increases in both ΔEPA and ΔDHA compared with fish oil. This suggests that for a similar dose of ω-3 FAs, the PL form results in higher circulating ω-3 FAs levels compared with TG form. Our results are consistent with the study by Ramprasath et al. [17], where krill oil resulted in 1.45-fold higher increases in ΔEPA +DHA than fish oil after 4 wk of supplementation. However, Yurko-Mauro et al. [35]

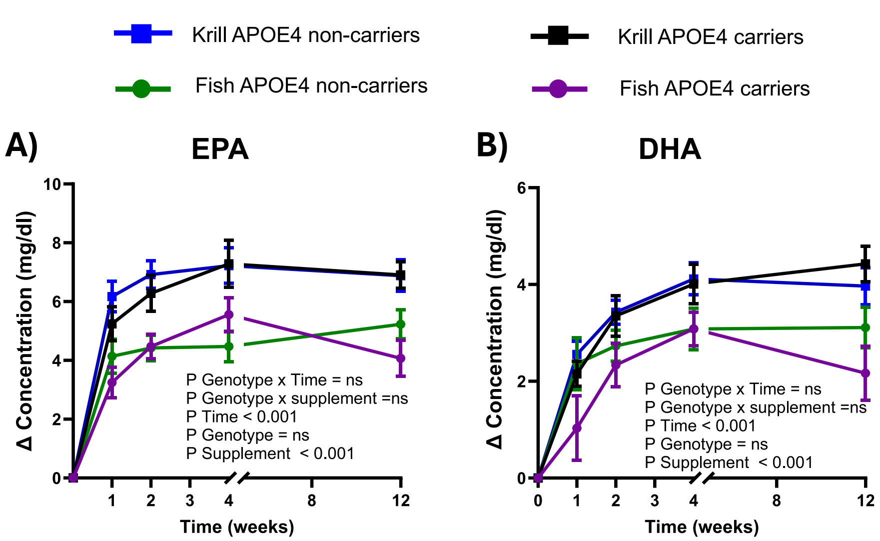

FIGURE 4. Plasma delta concentrations from baseline of (A) EPA and (B) DHA with Krill-oil supplement (black; APOE4 carriers, blue; APOE4 noncarriers) and fish-oil supplement (red; APOE4 carriers, green APOE4 noncarriers). Values are means ± SEM. Missing data were as follows: at week 4 (APOE4 carriers, $n = 2$, fish oil) and at week 12 (APOE3 carriers, krill oil, $n = 1$; fish oil, $n = 1$). APOE4, apolipoprotein E ε4.

found no difference at a similar dose (1.3 g/d EPA + DHA). The variability of krill oil composition might account for these differences. Commercial products contain 19% to 80% PLs [36,37]. In a study by Yurko-Mauro [35], capsules had 44% PLs without subclass specification, whereas ours contained 57% PL, mostly in PC form. The lack of detailed compositional reporting makes cross-trial comparisons difficult. In addition to EPA and DHA, other plasma FAs were examined to determine whether the interventions changed the plasma fatty acid composition. At baseline, total plasma fatty acid profiles did not differ between the krill oil and fish oil groups, confirming that randomization was effective with respect to overall fatty acid status. Furthermore, aside from EPA and DHA, no significant differences were detected between supplements when comparing changes from baseline to 12 wk, except for C18:1 ω-7 and C22:5 ω-3 (Supplementary Table 2). C22:5 ω-3 corresponds to docosapentaenoic acid (ω-3 DPA), an intermediate ω-3 FA within the EPA and DHA metabolic pathway. In humans, the conversion of EPA to DHA is quantitatively limited, and circulating DPA largely reflects EPA availability. The greater increase in DPA with krill oil is consistent with higher EPA concentrations. The higher plasma EPA and DHA found with krill oil supplementation are plausibly related to the chemical form of the ω-3 FAs, with fish oil providing them mainly as TG, whereas krill oil provides them mainly as PL-bound ω-3 FAs. The higher plasma EPA and DHA concentrations found with krill oil in this study are probably not related to higher absorption of EPA and DHA from krill oil supplements. The greater plasma ω-3 enrichment observed with krill oil may reflect differences in the digestive and postabsorptive handling of PL-bound compared with TG-bound fatty acids. Previous human studies using double-labeled PC have shown prolonged retention of PL-derived fatty acids and choline in plasma lipoproteins and erythrocytes, indicating distinct kinetics compared with TG-derived fatty acids [38]. Mechanistically, PC undergoes partial intestinal hydrolysis to lysophosphatidylcholine (LPC), which can be absorbed intact, reacylated, and incorporated into circulating lipoprotein PL [39]. This pathway may favor incorporation into slower-turnover lipid pools, contributing to higher and more sustained circulating ω-3 levels. Together, these human data provide a plausible biological basis for the enhanced plasma ω-3 response observed with krill oil supplementation. Moreover, Sung et al. [24] found that, after a single dose intake, krill oil ω-3 FAs were predominantly incorporated into plasma PLs species, including PC, LPC, and phosphatidylethanolamine, whereas those from fish were mainly distributed within neutral lipids as TG and diglycerides. Indeed, after ingestion, dietary PLs are hydrolyzed by pancreatic phospholipase A_2 into FFA and lysophospholipids (e.g., LPC), which are absorbed by enterocytes, re-esterified, and secreted in chylomicrons [40]. Once in plasma, chylomicrons exchange PL-bound ω-3 FAs with HDL, whereas ≤20% of PLs may be absorbed intact, and a fraction of LPC can reach the liver via the portal vein, potentially influencing hepatic redistribution of ω-3 FAs to peripheral tissues [41]. These different transport and incorporation pathways for PL ω-3 FAs contrast with the predominantly chylomicron-mediated transport of TG ω-3 FAs. This can, in part, explain the results found in this study. In addition, krill oil supplementation led to an increase in HDL-C concentrations after 12 wk (0.149 mmol/L, corresponding to an ~7.7% rise), whereas fish oil did not modify HDL-C levels (data not shown). This observation is relevant because HDL particles constitute a major circulating pool of PL and play a central role in lipid transport. An expansion of the HDL compartment may therefore have contributed to the greater enrichment of DHA within plasma PL in the Krill-oil group.

It is often argued that the higher postsupplementation concentrations of EPA and DHA after krill oil supplementation are mainly due to the PL form of ω-3 FAs. However, this explanation alone may not fully account for the observed differences relative to fish oil. Another biochemical feature of krill oil is its choline content. In a previous study on pregnant females [42], DHA supplementation was combined with different choline doses. Higher choline doses were found to increase plasma PC-DHA and erythrocyte total DHA by 35% and 10%, respectively. In this study, choline supplementation also significantly increased total plasma DHA concentrations. These findings indicate that higher choline availability facilitates DHA esterification and mobilization within plasma PL. Moreover, as a precursor of acetylcholine, choline may also support cognitive function, although this was not examined in the present study [43–45].

In this trial, krill oil resulted in higher ΔEPA and ΔDHA in plasma. However, it is important to differentiate between differences that are statistically significant and physiologically or clinically meaningful. Higher circulating ω-3 FAs concentrations may translate into improved clinical outcomes or also reflect reduced uptake by peripheral tissues. In fact, plasma enrichment in ω-3 FAs after supplementation is widely used in intervention studies, given its accessibility and responsiveness to the intake changes [46]. However, it represents only part of the picture, as circulating levels can fluctuate with diet, fasting state, and lipoprotein metabolism. Still, higher plasma EPA and DHA have been associated with lower risk of cognitive decline and cardiovascular disease [47,48]. In fact, animal studies have shown that when rats were orally gavaged with ^{3}H-labeled DHA esterified either to PC or to TG, the amount of radioactivity recovered in the cortical PL was 6 times higher with PC-DHA than in the esterified TG-DHA group. This suggests that increasing plasma ω-3 FAs in PL form may increase their delivery to the brain [49]. Together, these findings support the idea that circulating ω-3 FAs, despite their variability, can serve as a surrogate measure of their availability to target tissues. Another measure that is more reflective of EPA and DHA incorporation in tissues is the erythrocyte levels of EPA and DHA. Erythrocyte-EPA and DHA levels are a complementary and more stable measure, as they reflect long-term incorporation into cell membranes and correlate strongly with both plasma DHA and internal organ DHA status [46]. Therefore, it is important to analyze both plasma and erythrocyte fatty acids to obtain a more comprehensive picture of ω-3 FAs status and tissue incorporation. In this study, erythrocytes were also collected for future analyses that will explore tissue incorporation, reinforcing the findings of this study.

Results from this study also demonstrated that females had greater increases of ΔEPA in plasma than males; however, ΔDHA were not sex dependent. This aligns with previous findings from our group where females had higher EPA levels in the plasma PL after ω-3 FAs supplementation, whereas plasma DHA increases were not influenced by sex [14]. Notably, among females, those receiving krill oil had more pronounced ΔEPA than those on fish oil. In males, EPA responses were consistently lower regardless of the supplement type. These findings highlight that sex-specific physiology and supplement formulation both play critical roles in modulating ω-3 FAs concentrations in plasma.

In this study, no significant effect of APOE4 genotype was found. However, most APOE4 carriers were females (8 females, 2 males per group), which may have limited the detection of genotype-dependent differences. Previous work from our laboratory reported greater plasma EPA increases in APOE4 carriers, with DHA being less affected [14]. Other studies also describe altered ω-3 FA transport and metabolism in APOE4 carriers [13,50,51]. Taken together, although the present results did not reveal clear genotype effects, the limited sample size and sex imbalance among APOE4 carriers likely constrained our statistical power. Future analyses stratifying by both sex and APOE4 status will be essential to clarify the interplay between these factors in modulating ω-3 FAs concentrations after supplementation.

The present trial has several methodological strengths. A major strength is that the 2 supplements provided comparable doses of EPA and DHA. This allowed plasma ω-3 FAs comparison enrichment between the 2 supplements, differing only by the chemical form of the ω-3 FAs. Also, the inclusion of both males and females enabled the examination of sex-specific responses, which are usually overlooked in other studies comparing fish and krill supplementation. In this trial, the dietary EPA and DHA were also estimated using an FFQ. Dietary intake of ω-3 FAs was low and not statistically different between both supplements' groups. However, certain limitations must be acknowledged. First, the number of participants ($n = 72$) was lower than the original recruitment target ($n = 160$), which may have limited the detection of smaller effects, particularly in secondary outcomes such as APOE4 genotype. In addition, this limited the study's power to detect 3-way ANOVA interactions between sex, supplementation, and genotype. The study was designed to achieve 80% statistical power ($\beta = 0.20$) for the primary outcomes (plasma EPA and DHA concentrations). However, it was not specifically powered to detect 3-way interactions between sex, supplementation, and APOE4 genotype. As a result, 3-way ANOVA is exploratory and should be interpreted with caution. Although we aimed to recruit an equal number of males and females for this study, the final sample was imbalanced. This discrepancy is largely due to lower participation rates among males in clinical trials at our site, a trend we have consistently observed. Moreover, although participants were free of a diagnosed dementia or self-reported neurodegenerative disorder, the inclusion criteria did not exclude individuals with undiagnosed prodromal cognitive conditions such as amnestic or nonamnestic mild cognitive impairment. As a result, some participants may have been in the early stages of neurodegenerative processes, which may limit the interpretability and generalizability of the findings to strictly cognitively healthy populations. The present report focuses on plasma total fatty acids, which reflect short-term intake of lipids. However, erythrocytes were kept for complementary analysis. Finally, in this study, participants were healthy adults aged 30 to 65 y, free of major metabolic, inflammatory, or cognitive disorders. As such, the results are most applicable to generally healthy middle-aged adults. Extrapolation to older populations, individuals with chronic diseases, or those with cognitive impairment should not be made. Future studies targeting these specific populations will be necessary to determine whether these findings extend beyond the present cohort.

In conclusion, in this trial, krill oil resulted in higher ΔEPA and ΔDHA in plasma, indicating that the circulating levels of ω-3 FAs after supplementation are dependent on the chemical form of the supplements. These results mean that for a similar dose of ω-3 FAs, krill oil increases more EPA and DHA concentrations in males and females.

Acknowledgments

We would like to thank all the participants who contributed their time and effort to this study.

Author contributions

The authors' responsibilities were as follows – MP: designed research and primary responsible for final content; IL: conducted research and performed statistical analysis; AC: provided support with follow-up visits; IL, AV: analyzed data; IL, MP: wrote the paper; and all authors: read and approved the final manuscript.

Declaration of generative AI and AI-assisted techologies in the writing process

During the preparation of this work, the author(s) used Microsoft Copilot in order to syntax, orthograph and overall readability of the paper. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Conflict of interest

The authors report no conflicts of interest.

Funding

This work was funded by the Canadian Institutes of Health Research (201909PJ). The study funder had no role in the design, execution, or interpretation of the research.

Data availability

Data described in the manuscript, code book, and analytic code will be made available upon reasonable request to the corresponding author, pending approval for noncommercial academic use.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajcnut.2026.101346.

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